

Comparison of linear voice coil and reluctance actuators for high-precision applications

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Abstract—This paper concerns the comparison of three linear reluctance actuators and four voice coil actuators for high-precision applications. The short-stroke actuators with reasonably low variation of the force are optimized on actuator mass and power dissipation. For actuators with a significantly high force variation, the configuration is modified with the objective to improve the actuator properties for high-precision applications. Finally, the comparison is made on performance regarding, force, mass, losses and stiffness.

I. INTRODUCTION

In the semi-conductor industry, wafer scanners are developed to produce integrated circuits. In wafer scanners accurate positioning is applied, for which high-precision actuation is necessary. Such high-precision actuation is achieved in two actuator stages, a short-stroke stage stacked on a long-stroke stage, as illustrated in Fig. 1. Short-stroke actuators, [1] - [2] have a millimeter range with an accuracy of nanometers and long-stroke actuators, [1] - [5], normally have a range of meters with an accuracy of sub-micrometers.

In prior designs for high-precision actuators, the relatively low variation of the force, over a stroke of 1 mm, is the main reason to choose for voice coil actuators. In state-of-the-art wafer scanners different types of voice coils are applied for high-precision actuation. This can be gravity compensator's, linear and planar voice coil actuators, which have been investigated for many years, [1] - [7].

In the development of new wafer scanners the production speed gains importance. A higher throughput is achieved by increasing the acceleration of the wafer-stage. Therefore, the force is increased, which results in a larger actuator size and an increased moving mass. Moreover, the mass of the water cooling of voice coil actuators is significantly rising. To break through this spiral, the research focuses on mass-reduction and hence, reluctance actuators are investigated, which can achieve a higher force density than voice coil actuators.

In some papers comparisons are made between optimized actuators of a certain type. As for variable reluctance actuators [8], optimized for high performance, however, not for high-precision applications. Other papers compare lightweight voice coil actuators to prior designs [1]. However, the comparison between linear reluctance and voice coil actuators for high-precision applications is not investigated as such.

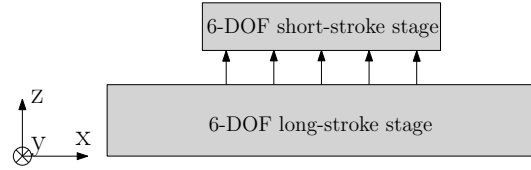


Fig. 1. Magnetically suspended short-stroke stage with multiple high-precision actuators, stacked on a magnetically levitated long-stroke stages, both with 6-DOF.

This paper concerns the comparison of four voice coil actuators to three types of reluctance actuators. Those actuators are designed for a static force in one direction. This force must be applied for a millimeter stroke in 6 DOF. The performance of those actuators is first evaluated on the force variation, which is highly important for high-precision applications. Then the most suitable configurations are optimized on power dissipation and actuator mass.

II. ACTUATOR TOPOLOGIES

The comparison of two different types of actuators must be carefully defined. Several criteria can be investigated for an actuator comparison, such as, actuator mass, force, power dissipation, force density, volume, position dependency of the force, manufacturability, etc. The comparison criteria are dependent on the prescribed specifications of the application. For high-precision applications all the mentioned criteria are important, however this paper concentrates on the power dissipation, the moving mass and the position dependency of the force.

The actuators are designed with the objective to achieve a force of 10 N in z -direction, for a stroke of 1 mm. For every actuator, the variation of the force with motion, is evaluated. The variation of the force is defined as the percentage of the force normalized to the driving force at the nominal position. The force variation is evaluated for a stroke of 1 mm in the x -, y - and z -direction and a rotation of ± 5 mrad. For the reluctance actuators with a force variation in the range of the currently used voice coil actuators, the dissipation related to the moving mass is compared. The moving mass is considered as the most important design constraint, because this contributes to the mass of the short-stroke wafer-stage.

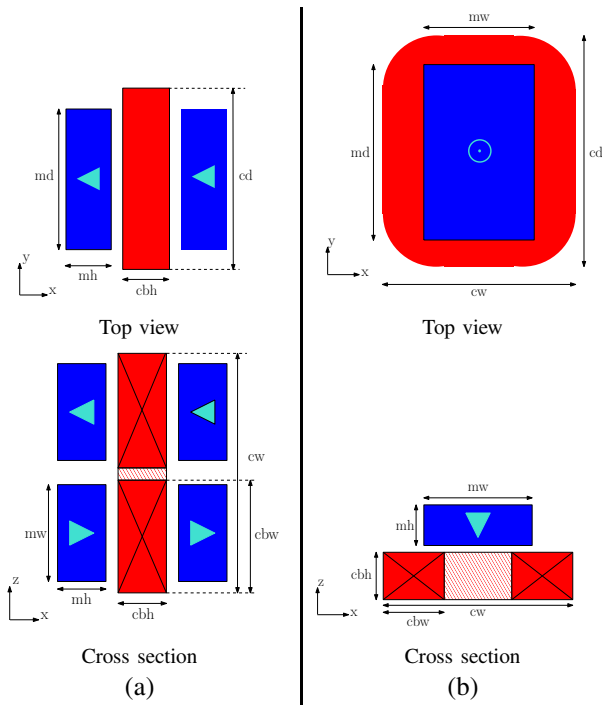


TABLE I
VOICE COIL ACTUATOR OPTIMIZATION PARAMETERS

In the comparison of voice coil and reluctance actuators, two types of iron less voice coil actuators have been considered, which consist of a stationary coil and one or four moving permanent magnets. The first type is a linear voice coil actuator with four moving magnets as shown in Fig. 2(a). This actuator is chosen regarding its low force variation, due to a nearly homogeneous magnetic flux density in the airgap. This voice coil actuator has a rectangular coil and magnets. The second type is a voice coil actuator with one moving magnet above a static coil as shown in Fig. 2(b). This voice coil actuator is evaluated, because a configuration with one magnet is easier to construct than a configuration with four moving magnets. Three variants of the configuration with one moving magnet are investigated. Namely, a square shaped, a round shaped and a rectangular shaped magnet and coil. The optimization parameters of the voice coil actuators are given in Table I.

TABLE II
RELUCTANCE ACTUATOR OPTIMIZATION PARAMETERS

reluctance actuator, which is the plunger type actuator [9] that can be designed to have a position independent force in the moving direction. This reluctance actuator can also not produce a bidirectional force. Contrary to the C-core actuator, the stator of the plunger type actuator does not have to be placed on top of the mover, but can be placed perpendicular to the driving direction. Fig. 3(c) shows a hybrid reluctance actuator [10] pre-biased with permanent magnets, able to achieve a linear force-current relation. Moreover, this actuator can produce a bidirectional force and the pre-biased magnets can apply a static force to compensate for gravity without static losses. The optimization parameters of the reluctance actuators are specified in Table II.

In this section, the force variation of four voice coil actuators is studied to be able to make a comparison to the evaluated reluctance actuators. The investigated voice coil actuators are first optimized for minimal power dissipation. Thereafter, the configurations with a moving mass of 25 grams are evaluated on the variation of the force.

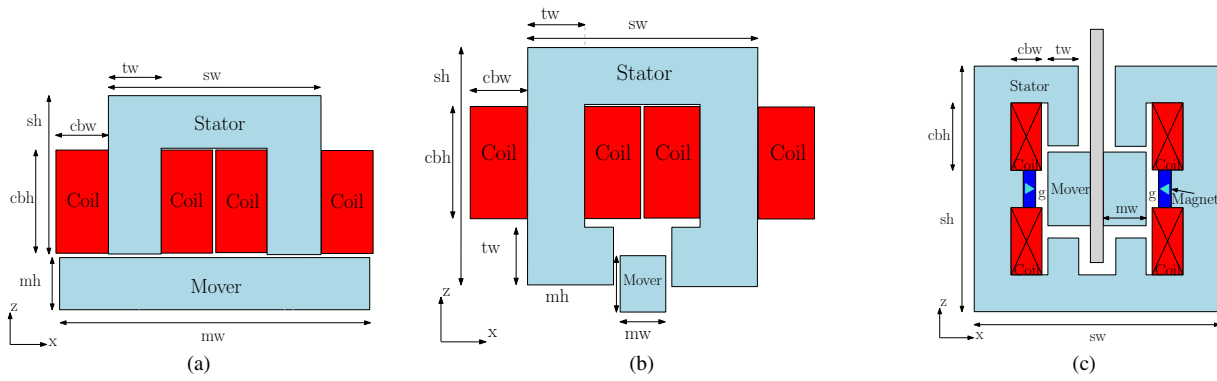


Fig. 3. Cross section of the investigated reluctance actuators, (a) C-core actuator, (b) plunger type actuator, (c) hybrid reluctance actuator.

integrating the Lorentz force over the whole coil. This results in a more accurate solution and can include position effects.

Voice coil actuator with four rectangular magnets

Fig. 2(a) shows a schematic drawing of the top view and the cross-section of the linear voice coil actuator with four moving magnets. The dimensions of this optimized voice coil actuator are given in Table III. For the optimization no back iron is included. The four moving magnets of this actuator have a significantly large depth (in the y -direction) compared to the height and width.

Fig. 4 show the force variation of this actuator with four moving magnets. The variation of the force is defined as the proportion of the force related to the nominal force in the driving direction, in percentages. Fig. 4(a) elucidates that the force variation of this voice coil with four permanent magnets is low, which is 1.6% in the driving direction. Fig. 4(b) and (d) show that the force variations in the z - and x -direction, due to rotation around the x - and y -axis are respectively 0.05% and 0.17%. The force variation in the y -direction due to rotation around x - and y -axis is 0.27%, which is higher than the force variation in the x -direction, because the actuator is larger in the y -direction. Fig. 4(c) shows that no force variation in the x -direction is present with translation in the xy -plane, as a result of the Lorentz force that only acts in the direction perpendicular to the magnetic flux vector and the vector of the current. For the same reason no force variation in the y -direction is present. The optimization results on minimal power dissipation for the different moving masses are given in section V.

B. Voice coil actuator with one square-shaped moving magnet

Fig. 2(b) shows a top view and cross-section of a square-shaped voice coil actuator with one moving magnet. The exact dimensions of the optimized voice coil actuator are given in Table III.

Fig. 5 shows the position dependency of the force of the optimized square-shaped voice coil actuator. Fig. 5(a) shows the significantly higher force variation (8%) in the driving direction compared to the voice coil actuator with four magnets (1.6%). This is a result of the less

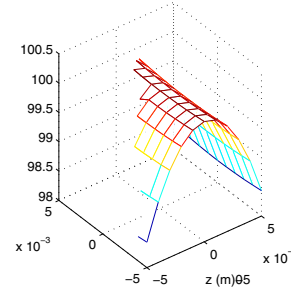


TABLE III
DIMENSIONS (mm) OF THE VOICE COIL ACTUATORS OPTIMIZED FOR A MOVING MASS OF 25 GRAMS

Voice coil actuator type	Coil bundle width (cbw)	Coil bundle height (cbh)	Coil width (cw)	Coil depth (cd)	Magnet width (mw)	Magnet height (mh)	Magnet depth (md)
Four rectangular magnets	5.7	3.6	13.6	72.8	5.4	2.539	59.2
Square-shaped - one magnet	11.2	8.4	34.8	34.8	21.1	7.2	21.1
Axis-symmetric - one magnet	15	9.6	33.8	33.8	23.6	7.8	23.6
Rectangular - one magnet	8.9	6.3	21.5	75.8	9.5	5.12	65.9

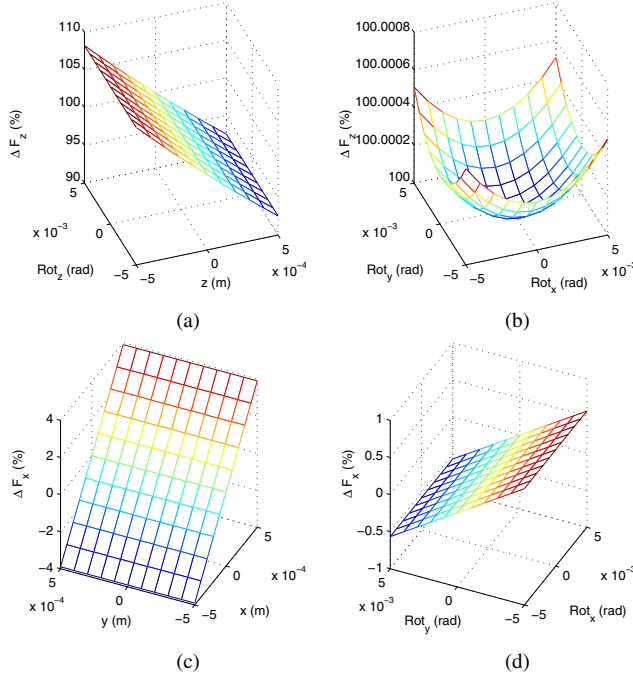


Fig. 5. Force variation of the square-shaped voice coil actuator, (a)-(b) in the actuation direction and (c)-(d) perpendicular to the actuation direction, with translations and rotations in the various directions.

is 0.6%. This is larger than for the voice coil actuator with four moving magnets (0.17%).

C. Voice coil actuator with one axis-symmetric moving magnet

An axis-symmetric voice coil actuator has a similar cross-section as a square-shaped voice coil as shown in Fig. 2(b). The difference with the square-shaped voice coil actuator is the top view, which is circular. The exact dimensions of the optimized axis-symmetric voice coil actuator are given in Table III. For the given dimensions, the diameter of the circular shape is intended.

The force variation of the axis-symmetric voice coil actuator is comparable to the square-shaped actuator as shown in Fig. 5. The optimization results on power dissipation and moving mass, compared to the other evaluated actuators are shown in section V.

D. Voice coil actuator with one rectangular moving magnet

Fig. 2(b) shows a top view and cross-section of a rectangular voice coil actuator. However, the optimized rectangular voice coil actuator is significantly larger in the y -direction compared to the width and height in

respectively the x - and y -direction. The exact dimensions of the optimized rectangular voice coil actuator are given in Table III.

For this actuator the force variation in the x -direction is 9%, which is larger than for the square-shaped voice coil actuator (4%). However, the force variation in the y -direction is 1.3%, which is even lower than for the voice coil actuator with four moving magnets (1.6%). This difference in both directions is a direct consequence of the rectangular shape.

IV. RELUCTANCE ACTUATORS

The reason that reluctance actuators are not applied for high-precision actuation in previous applications, is mainly the high force variation and the nonlinear current-force relation. Both properties are highly dependent on the design of reluctance actuators as shown in next section. Moreover, the high force variation in the driving direction has been a major difficulty in prior considerations, but the control bandwidth is nowadays significantly higher, due to the increased calculation capacity of available processors.

Other foreseen difficulties with reluctance actuators compared to voice coil actuators are magnetic hysteresis and eddy current effects. Eddy currents are neglected for this static analysis and the hysteresis effects are limited, because a low hysteric core material is applied for the reluctance actuators. The low hysteric material is Vacoflux 50 with a coercive magnetic field strength of $H_c = \pm 30 A/m$. Therefore, no hysteresis effects are taken into account in the comparison between the reluctance and voice coil actuators.

Reluctance actuators can be analyzed with several methods. The most commonly used methods are magnetic equivalent circuits (MEC) and the finite element method (FEM). Magnetic equivalent circuits are mainly suitable for initial sizing of the actuators, because leakage fluxes and fringing effects are difficult to model with equivalent reluctances. Although there exist analytical descriptions of airgap permeances [16], this paper practices 2D MEC-models without leakage fluxes and fringing effects, to achieve quickly an initial design. The finite element (FEM) package FLUX2D [15] is used to optimize the reluctance actuator designs and to investigate the position dependency of the force. The results of the simulations on the force variation are given for the three evaluated reluctance actuators in next sections.

A. C-core reluctance actuator

Fig. 3(a) shows a drawing of the C-core actuator from which the exact dimensions for the evaluation are given

TABLE IV
DIMENSIONS (mm) OF THE RELUCTANCE ACTUATORS EVALUATED ON THE VARIATION OF THE FORCE

Reluctance actuator type	Coil bundle width (<i>cbw</i>)	Coil bundle height (<i>cbh</i>)	Tooth width (<i>tw</i>)	Mover width (<i>mw</i>)	Mover height (<i>mh</i>)	Stator width (<i>sw</i>)	Stator height (<i>sh</i>)	Actuator Depth
C-core actuator	13.5	27	3.5	47	3.5	36	32.5	3.5
Plunger type actuator	6.25	12.5	6.25	5	6.25	25	26	25
Hybrid (pre-bias) actuator	3	6	3.5	4.5	8	25	25	25

in Table IV. This C-core reluctance actuator is evaluated on the variation of the force as shown in Fig. 6.

Fig. 6(a) shows the force variation in the driving direction with translation in the driving direction. This force variation is quadratic related to the position, and is therefore significantly higher than for the evaluated voice coil actuators. Nevertheless, the force in the driving direction will be controlled. Therefore, the force in the driving direction will not contribute to cross-talk effects for other actuators that are placed to apply a force in other directions. The current-force relation of a C-core actuator is also quadratic, however this relationship can be linearized, for a small current variation, with a stator and coil segment on both sides of the mover. The coil current in both coil segments is then manipulated to linearize the current-force relation, in the same manner as is shown for the hybrid reluctance actuator in Fig. 10.

Fig. 6(b) shows the force variation in the driving direction with rotation around the y -axis, which is 0.18%. This is slightly higher compared to the force variation for the voice coil actuator with four moving magnets (0.06%). Fig. 6(c) show the force variation in the direction perpendicular to the driving direction for different mover sizes. The mover width is enlarged from the width of the stator (sw) to the stator width plus 11 times the airgap length (g). The force variation in the perpendicular direction is less than 0.3% for a mover width of $sw + 5g$. This is significantly lower than the 4% force variation of the square-shaped voice coil actuator as shown in Fig. 5(c). However, the voice coil actuator with four moving magnets has no variation of the force as shown in Fig. 4(c). Fig. 6(d) shows the rotational variation of the force of the C-core actuator, which is in the same order as for the square-shaped voice coil actuator, namely below 0.6%.

Reduction of the force variation by increasing the mover width of a C-core actuator is investigated more in detail over a stroke of 10 mm as shown in Fig. 7. Instead of the variation of the force, the stiffness (N/m) in the x -direction is evaluated and the values of the stiffness at the position at which the mover is centered, are given in Table V. The mover width is enlarged with 11 times the airgap length (g) compared to the stator width (sw). In this way the stiffness in the x -direction is reduced from 825 to 19 N/m. The same stiffness reduction can be achieved for the force variation in the y -direction by enlarging the mover depth. This way the force in the driving direction is practically decoupled from the position in the xy -plane for a stroke of 1 mm. The force variation in the direction perpendicular to the driving direction, is hence similar

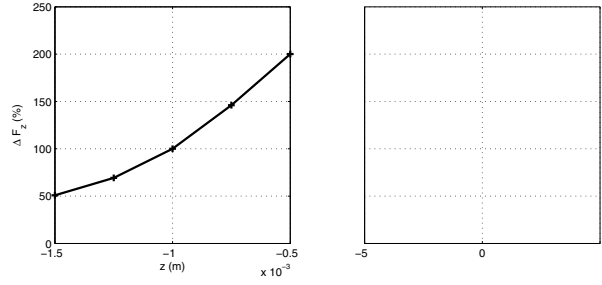


TABLE V
STIFFNESS OF THE C-CORE ACTUATOR WITH VARIABLE MOVER
WIDTH (sw = STATOR WIDTH AND g = AIRGAP LENGTH)

Mover width	Stiffness [N/m]
sw	-825
$sw + g$	-629
$sw + 3g$	-234
$sw + 5g$	-99
$sw + 7g$	-51
$sw + 9g$	-33
$sw + 11g$	-19

TABLE VI
C-CORE ACTUATOR OPTIMIZATION PROPORTIONS

Parameter	Symbol	Value
Stator width	sw	$mw - 11g$
Stator height	sh	$sw - tw$
Coil height	ch	$sw - tw - 2mm$
Coil width	cw	$ch/2$
Coil length	cl	$4tw + \pi cw$

optimize the mover of the C-core actuator is the mover width. Enlarging the mover width results in an increased stator size, which enlarges the winding area of the coil, and hence decreases the dissipation. Consequently, no optimum can be found for the optimization of the C-core actuator on dissipation and mover width. Therefore, the C-core based on the design in Fig. 3(a) is first evaluated for different current densities in the coils and different values of the magnetic flux density in the iron and airgap, with the magnetic equivalent circuit (MEC) model. The designs are achieved with the MEC-model without leakage and fringing and the parameters and variables are given in respectively Tables II and VI. Fig. 8 is constructed in a number of steps. First, the total mass of the C-core actuator is simulated for several fixed current densities and variable magnetic flux densities in the airgap. This results in a C-core actuator optimization on total mass for each current density. Thereafter, the minimum of those curves is approached for a fixed magnetic flux density of 0.4 T. Moreover, the upper boundary is plotted with a fixed flux density of 1 T.

Four C-cores actuators optimized on total mass, with a flux density of 0.4 T, are simulated with the FEM package, FLUX2D, to verify the MEC-results. The moving mass of the four verified actuators is 25, 30, 35 and 43 grams. The dissipation in the MEC models is determined to be between 10 and 15% higher than the dissipation in the FEM simulations, due to the neglected leakage and fringing in the MEC models. The dimensions of the four simulated C-core actuators are given in Table VII.

B. Plunger type reluctance actuator

The plunger type reluctance actuator, as shown in Fig. 3(b), is evaluated because of its position independent force, which is caused by the fact that not the airgap length varies in the main flux path, but the area of the main flux path. Consequently, the inductance varies linearly with the mover position and, hence, the force is position independent. Moreover, the stator part can

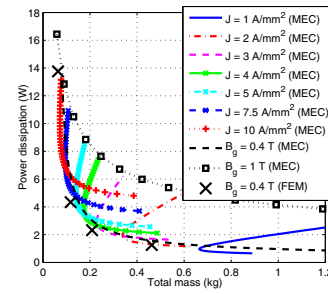


Fig. 8. C-core reluctance actuator evaluated on total mass and dissipation. The MEC results are verified by FEM simulations.

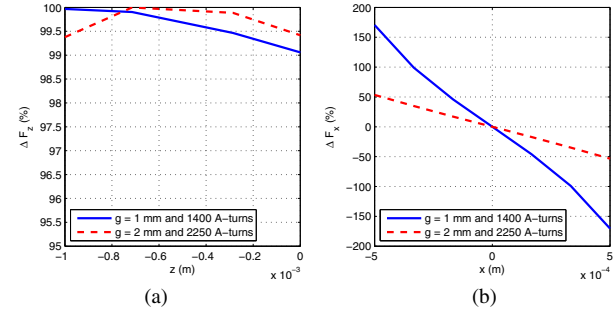


Fig. 9. Force variation of the plunger type reluctance actuator, in the (a) actuation direction, and (b) perpendicular to the actuation direction.

be placed perpendicular to the driving direction, which makes it easier to mount the mover to the short-stroke stage. The exact dimensions of the evaluated plunger type actuator are given in Table IV.

The force variation of this actuator is shown in Fig. 9. Fig. 9(a) shows the force variation in the driving direction, which is below 1% for a stroke of 1 mm. Fig. 9(b) shows the force variation perpendicular to the driving direction, which is reduced by increasing the airgap between the stator and plunger from 1 to 2 mm. This modification reduces the force variation from 170 to 54%. The current is increased from 1400 to 2250 A-turns to achieve a force of 10 N, with an airgap of 2 mm. For the configuration used in the FEM simulations the dissipation in the coils is 25.3 W and 65.4 W for respectively an airgap of 1 and 2 mm. Increasing the airgap length reduces the force variation, but the dissipation increases more than 2 times. Nevertheless, the force variation is still more than 50%, which is significantly higher than the variation of the force of the previously evaluated actuators. Hence, this plunger type actuator is not optimized further for the actuator comparison.

C. Hybrid reluctance actuator

In contradiction to the C-core and the plunger type actuator, the hybrid reluctance actuator [10], as shown in Fig. 3(c) is able to produce a bidirectional force. The dimensions of the evaluated hybrid reluctance actuator are given in Table IV. The actuator is hybrid because of the permanent magnets, used to pre-bias the actuator. Besides a bidirectional force, a linear force-current relation can

TABLE VII
DIMENSIONS (mm) OF THE EVALUATED C-CORE RELUCTANCE ACTUATORS IN FLUX2D, OPTIMIZED FOR A MAGNETIC FLUX DENSITY OF 0.4 T

Actuator moving mass	Total mass (g)	B_{max} (T)	Power dissipation (W)	Coil bundle width	Coil bundle height	Tooth width	Mover width	Mover height	Stator width	Stator height	Actuator depth
25 grams	64	1.25	13.76	6.4	8	8.87	39.2	8.87	28.2	18.87	8.87
30 grams	209	1.26	4.34	7.95	15.9	8.87	47	8.87	36	26.77	8.87
35 grams	225	1.28	2.32	11.85	23.7	8.87	54.9	8.87	43.9	34.57	8.87
43 grams	462	1.31	1.26	18.1	36.2	8.87	67.4	8.87	56.4	47.07	8.87

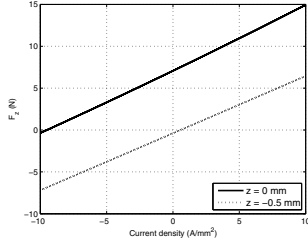


Fig. 10. Linearized force-current relationship of the hybrid reluctance actuator for two different mover positions.

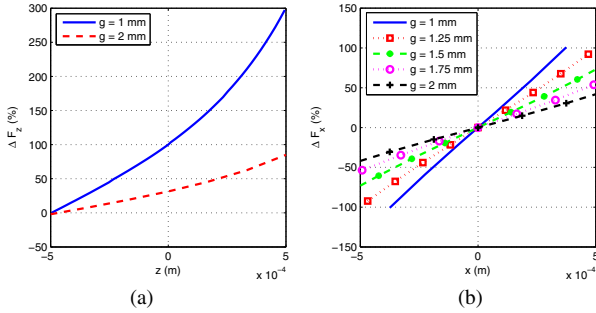


Fig. 11. Force variation of the hybrid reluctance actuator for different airgap sizes, in the (a) actuation direction, and (b) perpendicular to the actuation direction.

be achieved. Fig. 10 shows a linear force-current relation when the upper and lower coils are excited simultaneously. Moreover, the lower airgap is larger than the upper airgap to achieve a static force to compensate for the gravitational force with minimal dissipation.

The force variation in the direction perpendicular to the driving direction is comparable to the force variation of the plunger type actuator. Similar to the plunger type actuator, the stiffness can be reduced by increasing the airgap (g) between the mover and the magnets. Fig. 11(a) shows the decreased force variation in the driving direction, with an airgap between the magnet and mover of 1 and 2 mm. Increasing the airgap, reduces the force variation, but decreases also the static force more than 3 times. Fig. 11(b) shows the force variation in the perpendicular direction with an increased airgap between the stator and mover. The force variation in the direction perpendicular to the driving direction is reduced from 135% to 47%. This force variation is still significantly higher than the force variation of the evaluated voice coil and C-core actuators. Therefore, the stator tooth and mover width are modified to decrease the stiffness in the perpendicular direction.

Fig. 12(a) shows the hybrid reluctance actuator with

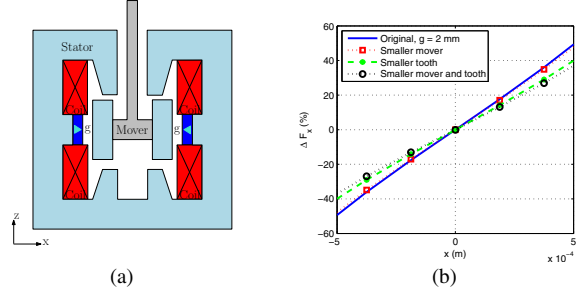


Fig. 12. (a) Hybrid reluctance actuator with modified stator tooth and mover width and the (b) force variation in the x -direction with modification of the stator tooth and mover width.

modified stator tooth and decreased mover size. The reduction of the stiffness is shown in Fig. 12(b). Now the total force variation is reduced from 135% to 37%, which is still undesirably high. Hence, this reluctance actuator is not considered for the actuator comparison further.

V. ACTUATOR MASS COMPARISON

Reduction of the mass of the short-stroke wafer-stage is extremely important to increase the production speed of integrated circuits. Therefore, the moving mass and minimal power dissipation is compared. From the three investigated reluctance actuators, only the evaluated C-core actuator is considered, because of the low variation of the force.

Recapitulatory, the actuators optimized on moving mass and power dissipation are:

- the voice coil actuator with four moving magnets,
- the square-shaped voice coil actuator with one moving magnet,
- the axis-symmetric voice coil actuator with one moving magnet,
- the rectangular voice coil actuator with one moving magnet,
- the C-core reluctance actuator.

Fig. 13, shows the comparison on moving mass between the optimized voice coil actuators and the C-core reluctance actuator. The C-core actuator was first optimized on total mass as shown in Fig. 8. The simulations with the optimization on total mass resulted in the configuration optimized for a magnetic flux density in the airgap of 0.4 T. When the moving mass is considered, as shown in Fig. 13, the optimal configuration is the one optimized for a magnetic flux density of 1 T. Configurations optimized for a flux density of 1 T result in very small stator tooth cross sections compared to the total actuator

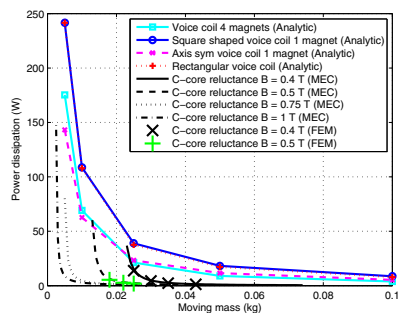


Fig. 13. C-core reluctance actuator compared to the voice coil actuator on moving mass and dissipation. MEC results are verified by FEM.

size. Consequently, a large winding area can be filled, which results in a low dissipation, but a significantly high coil mass. Therefore, the balancing between mover mass and stator mass is highly important in the C-core actuator design. Moreover, a magnetic flux density of 1 T in the airgap results in a flux density of 2.1 T in the core material. Hence, only cobalt-iron materials can be applied for the core material.

The power dissipation of an optimized C-core actuator for a mover mass of 25 grams and 1 T is 0.9 W. For a magnetic flux density of 0.5 T the dissipation is 1.9 W. This is much lower than for the voice coil actuator with four magnets, which has a dissipation of 21.3 W for a moving mass of 25 grams. This is more than 10 times higher than for the evaluated C-core actuator. The axis-symmetric voice coil actuators has a slightly higher dissipation (23.3 W) compared to the voice coil with four magnets. The square-shaped and rectangular voice coil actuators have a power dissipation of 38.9 W and 37.6 W respectively, which is almost two times higher than for the voice coil actuator with four magnets. Moreover, Fig. 13 shows that for higher mover masses of the C-core actuator, the magnetic flux density is not limiting, and hence, different materials can be chosen.

VI. CONCLUSIONS

This paper presents an evaluation of the comparison of linear reluctance and voice coil actuators for short-stroke high-precision applications. Seven different actuators are studied on force variation. Four voice coil actuators, namely, one with four moving magnets, a square-shaped actuator with one moving magnet, a axis-symmetric actuator with one moving magnet and a rectangular actuator with one moving magnet. Moreover, three reluctance actuators are studied, namely, a C-core actuator, a plunger type actuator and a hybrid (pre-biased) actuator.

The results of the evaluated C-core reluctance actuators on power dissipation and moving mass, show that reluctance actuators can achieve a 10 times higher force density compared to the evaluated voice coil actuators. The variation of the force of the voice coil actuator with four moving magnets is significantly lower than for the other three evaluated voice coil actuators with one moving magnet. There is even no force variation in the x - and y -direction, except for a rotation around respectively the

y - and x -axis, which is still 4 times lower than for the voice coils with one moving magnet. The variation of the force of reluctance actuators is strongly dependent on the configuration of the actuator. The variation of the force of a C-core reluctance actuator in the actuation direction is significantly higher than for the voice coil actuators. Therefore, this requires a different control strategy. For a C-core actuator the force variation in the direction perpendicular to the actuation direction and the rotational variation of the force is in the same order of magnitude as for the evaluated voice coil actuators, when the mover is sufficiently larger than the stator.

The plunger type reluctance actuator and the hybrid reluctance actuator as studied in this paper, have positive properties for the current and for the force in the driving direction. Although, the investigated actuators have a 200 times higher force variation in the direction perpendicular to the actuation direction compared to C-core actuators.

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